

One decision is firm. One is still open.

DECISION A · WEEKS · FIRM · FUND NOW

Bravo runs the **last public build** of two products that are no longer supported. Someone has already run their own code on its production engine. A free fork of the engine, plus Spring Boot 4, fixes both problems for **18 to 33 engineer-days**.

DECISION B · MONTHS · OPEN · THIS PACK DELIBERATELY DOES NOT CHOOSE

Upgrade, Temporal, or LORA. Three right answers to three different questions — and the one measurement that would settle the biggest number has not been taken.

Bravo is not approaching two end-of-support dates. It has passed both.

And it is running the **last build either product will ever release for free**. There is no further patch to take.

COMPONENT	BRAVO RUNS	SUPPORT ENDED
Camunda 7 Community Edition	7.23.0	14 Oct 2025 – the line is closed, the final release was 7.24.0, the GitHub repo is archived, and there will be no security patches ever again
Spring Boot 3.5.x (open source)	3.5.16 – the last free patch, 25 Jun 2026	30 Jun 2026
Java 17 (Oracle premier support)	17, on distroless and Temurin	30 Sep 2026 – 20 days away
Camunda 7.23 (Enterprise maintenance)	– we hold no licence –	13 Oct 2026 – 33 days away

1 · THERE IS SOMEWHERE SUPPORTED TO GO, AND IT IS CHEAP

Two community forks under the Apache 2.0 licence — **Operaton 2.1.4** and **CIB seven 2.2.0** — run on Spring Boot 4, leave the `ACT_` database tables unchanged, still accept the old `camunda: BPMN` namespace, and ship automated migration scripts. What ended is *Camunda's ownership of the project*, not the engine itself.

2 · SPRING BOOT FORCES THE DECISION, NOT CAMUNDA

Camunda 7.23 will not run on Spring Boot 4, and the dependency runs one way only. Moving to Boot 4 **forces** a decision about the engine; changing the engine does not force Boot 4. So “upgrade Spring Boot first and decide about the engine later” **is not an option**.

3 · THERE IS NO SECOND LINE OF DEFENCE

61 process definitions designed to run arbitrary code are sitting in the production engine, and **one of them has been confirmed to run** inside the production pod. An engine with no way to receive patches, and a deployment endpoint anyone can reach, is the worst combination on this list.

There are two decisions here, and they have different deadlines

Separating them is the single most useful thing this pack does.

DECISION A – URGENT		DECISION B – STRATEGIC
The question	How do we make Bravo safe to run while we take Decision B?	What platform should Bravo be on for the next 5 to 10 years ?
Deadline	Weeks	Months
The answer today	Firm. Fork the engine and upgrade Spring Boot in one change – 18 to 33 engineer-days, Rp45–125M, and no licence to buy . Then fix the code-execution hole and the database version-log drift.	Open. Option 1 as the destination, Option 2 (Temporal), or Option 3 (LORA).
Does it depend on the other?	No	No
What kind of question is it?	Technical, and it has a fixed end date	Not mainly technical – it is about which cost the organisation would rather carry for five years

Decision A should not wait for Decision B.

Every strategic path leaves Bravo running the retail book for **at least 12 months**: Temporal takes 12 to 18 months, LORA 15 to 24. And at 18 to 33 days, Decision A does more than hold the line. It puts Bravo on a **fully supported engine and a supported Spring Boot outright**. The Spring Boot half, 12 to 20 of those days, has to be paid under Option 2 anyway.

The three options, on the numbers that decide

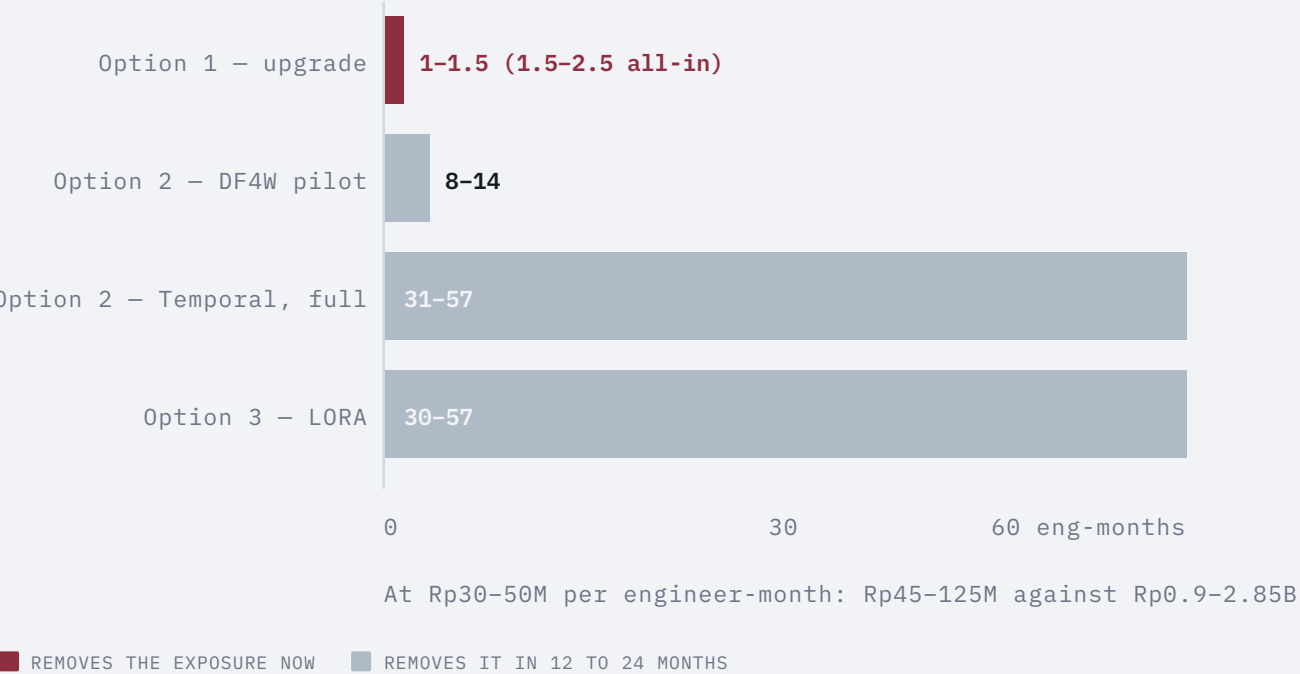
OPTION 1 – UPGRADE IN PLACE		OPTION 2 – TEMPORAL	OPTION 3 – LORA
What it is	Swap Camunda 7.23 CE for an Apache 2.0 fork and upgrade Boot 3.5.16 to 4.0.x, in one change. An automated OpenRewrite script does most of it, and the 53 BPMN files, the ACT_ tables and Java 17 all stay as they are	Rewrite about 60 BPMN definitions as Temporal Java SDK code. The 225 <code>JavaDelegate</code> classes become Activities. The other 94% of the service does not move	Close the gap in what LORA covers, prove it matches Bravo, move the rest of the book across, and switch Bravo off
Does the way we build change?	No	No – still step-by-step code, in the same language	Yes – data-centric GSM, in Go, with a custom SDK and planner
Effort	About 1 to 1.5 engineer-months to build (18 to 33 engineer-days); 1.5 to 2.5 including review and soak	31 to 57 engineer-months (or 8 to 14 for a DF4W-only pilot)	30 to 57 engineer-months (some of which overlaps LORA’s existing roadmap)
Elapsed time	1 to 2 months , one engineer	12 to 18 months, 5 to 8 engineers	15 to 24 months
Indicative one-off cost (at Rp30–50M per engineer-month)	Rp45M to Rp125M	Rp930M to Rp2.85B	Rp900M to Rp2.85B
New licences to buy	None – both forks are Apache 2.0	None – Temporal is already contracted, at Rp140M a month	None
Change to the monthly bill	None. The tier stays at about Rp58M a month in production – the cheapest of the three	About the same. Actions add Rp4.5–12M a month, probably inside the existing commitment	About Rp58M a month less in production (Rp70–100M less including test and staging), but you also staff a second platform
Effort is in engineer-months of an engineer who already knows the codebase, including their own testing but not UAT run by the business. Option 1’s figures are the Bravo team’s own , from their Camunda 7 Exit Plan, in engineer-days, at the confidence they stated themselves: “medium”. Ranges are low to high. They are not confidence intervals.			

The same three options, on the questions cost does not answer

	OPTION 1 – UPGRADE IN PLACE	OPTION 2 – TEMPORAL	OPTION 3 – LORA
How long it buys	A supported engine and a supported Spring Boot, on 6-month support lines (Operaton) or a contract you can buy (CIB seven)	Permanent – no workflow-engine vendor at all	Permanent – the platform stops existing
Does it fix the end-of-support problem?	Yes, completely – and roughly ten times faster than anything else here	Yes	Yes, but only after 15 to 24 months
New business capability	None	None	You can add an automated step without editing the workflow – proven across 172 activities with only 3 hard ordering constraints
Does it fix the architectural findings?	No	No – a faithful port carries the problems across with it	Yes, by replacing them
Keeps what Bravo does well	All of it – readable BPMN, Cockpit, questions across all loans in SQL, a durable row per reprocess attempt, failures that are bounded and classified	Mostly. Only 0.4% of the code touches the engine, there is no data migration and no retraining. Losing Cockpit is the real cost	No – it is replaced by a different model with different strengths
Do we still run two platforms afterwards?	Yes	Yes	No
Biggest risk	The database version log may not match the schema, which could stop the engine starting at cutover. That is diagnosable in hours. And a 6-month support line is a dependency we would be <i>choosing</i>	A large investment whose value depends on Bravo having a long life ahead of it	The cost of adopting a new way of building, plus LORA’s measured reliability gap at three times the volume

Option 1 is not a cheaper version of the others. It is a different scale entirely.

ONE-OFF ENGINEERING EFFORT, IN ENGINEER-MONTHS



20–40_x

MORE EXPENSIVE TO REACH A SIMILAR RUNTIME POSITION THROUGH OPTION 2

12–20

OF OPTION 1’S DAYS ARE THE SPRING BOOT HALF – WHICH OPTION 2 HAS TO PAY ANYWAY

WHY THE NUMBER COLLAPSED

An earlier version of this pack costed Option 1 at **11 to 19 engineer-months**, and assumed a Camunda **Enterprise licence** was needed for Spring Boot 4. Both were wrong. The Bravo team’s own *Camunda 7 Exit Plan* established the Apache 2.0 forks, and the figures here are **the team’s own estimate**, in engineer-days. We re-checked them against the working tree on 2026-09-10, and the inventory counts matched exactly.

WHAT OPTION 2 ACTUALLY BUYS FOR ITS 31 TO 57 MONTHS

Not a cheaper migration — it is **not meaningfully cheaper than Option 3**. What the reuse buys is **predictability, and the ability to go back**: only 2,004 of 497,970 lines (0.40%) touch a Camunda API, the 238 entities and 271 tables do not move, and a rollback lands on the same database.

Each option is the right answer to a different question

The fastest route to a decision is not to argue about the options. It is to work out **which question BFI is actually asking**.

IF WHAT YOU BELIEVE IS...	THEN THE ANSWER IS	BECAUSE
"Bravo should stay a strategic platform"	● OPTION 1 then reassess	For 18 to 33 days it lands on a maintained engine and a current Spring Boot. Option 2 costs 20 to 40 times more to reach a similar place. What Option 2 uniquely does is remove the engine dependency altogether; the fork route does not do that, it changes who maintains it
"Bravo has a limited life, but we have not decided what replaces it"	● OPTION 1 NOW decide later	18 to 33 days removes the exposure outright rather than postponing it, and commits you to nothing . It is the only option that keeps every choice open
"We want one LOS, and the GSM capability is worth what it costs to adopt"	● OPTION 3	It is the only option that ends with one platform . Sequence it so DF4W proves the approach on 5.5% of volume before the retail book moves
"We want one LOS, but GSM is the wrong model for us"	● OPTION 2, FULL SCOPE then consolidate onto it	Temporal is the execution layer both platforms would share. Nobody has costed this option, and it deserves to be costed
"Risk and compliance will not accept an unpatched runtime for any length of time"	● OPTION 1, PATH B whatever B turns out to be	It is the only route that ends on a supported engine <i>and</i> a supported Spring Boot. If a support contract is mandatory, that choice picks CIB seven over Operaton

What each option is good at — and what it leaves unanswered

OPTION 1 • UPGRADE IN PLACE Avoids disruption — and now it is the fastest and cheapest as well — Changes nothing about how an engineer or an analyst works. No data migration, no new way of building, no retraining, no new language — Keeps everything the comparison found Bravo genuinely does better: failures that are bounded and classified, with a designed place for the ones that cannot be handled; questions across all loans in SQL; a durable row for every reprocess attempt; a BPMN model an analyst can read; skills that are easy to hire — The cheapest workflow tier of the three, at about Rp58M a month — What it does not do: anything about Bravo’s architecture — five monoliths, the product identified in five places, four status vocabularies	OPTION 2 • TEMPORAL A permanent fix without changing how we build — and migration risk — The only option that removes the vendor dependency for good while keeping the step-by-step model the team already thinks in, on infrastructure BFI already owns and staffs — You can argue it matches Bravo element by element, rather than having to demonstrate every path. And a rollback lands on the same database — The workflow stays readable, and a diagram diff between what was generated and what was intended makes that checkable — Its weakness: it changes the engine and keeps the architecture exactly as it is. If the real complaint is Bravo’s <i>design</i>, this does not answer it	OPTION 3 • LORA One platform, and one new capability — Adding an automated step without editing a workflow model is real, it is proven, and it is the clearest thing LORA does better — The only option that ends the two-platform estate — and it removes about Rp58M a month of production workflow cost — Reliability currently favours LORA: staff step in on about 0.13% of applications, against Bravo’s 0.44% — Its price: a change in how everyone builds, whose cost is documented, partly fixable, and not fixed yet
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The objection to LORA’s way of building, weighed — neither dismissed nor swallowed whole

VERSION OF THE OBJECTION	STATUS	COST TO FIX
“Nobody can see what a product actually runs”	Fixable – generate an index of status by activity by product; the depchain-generator is about 80% built and not in CI	1–2 engineer-months for all three
“New engineers cannot learn it”	Fixable – write a week-1 curriculum	
“Nobody wrote down why it was designed this way”	Fixable – write the ADRs	
“A shared document and a shared planner leave no natural boundary for splitting the underwriting and surveyor teams”	Confirmed, and structural – and as of 2026-09-10 we have measured it on both sides	Not costed organisational

THE MEASURED HALF, AND WHY IT IS SHARPER THAN IT LOOKS

The boundary LORA lacks is one **Bravo is actively running**: two squads (BLCS and LN), two Jira boards, three operator consoles and separate sprint regression, with **no links at all between the two projects**. So Option 3’s organisational cost is concrete: **merge two working squads and re-cut them by product family**. It is not an abstract constraint.

THE FAIR SUMMARY

The objection is **real, mostly fixable, and cheap to fix compared with the migration** — but nothing has been fixed yet, and a decision that assumes it away will meet it at full strength during cutover.

It is a reason to **sequence Option 3 carefully**, and a legitimate reason to prefer Option 2 if the organisation values the current way of thinking highly. On this evidence it is **not** a reason to rule Option 3 out.

Four gaps stand between this pack and a confident answer on Decision B

All of them can be closed in **weeks**. One was closed on 2026-09-10, and we have kept it here with its answer, because that answer came with a caveat that is itself a smaller open question.

OPEN QUESTION	WHAT IT DECIDES	HOW TO CLOSE IT
Do we need a support contract for the workflow engine – and can we buy one for a BFI entity? (this replaces the old question about Camunda Enterprise pricing; no licence is needed)	It decides which fork , not which option. Yes → CIB seven : company-backed, cheaper migration, two ex-Camunda core engineers. No → Operaton : nine times the commit volume, 52 contributors	Architect and procurement. Start the CIB seven enquiry before finalising anything – its terms are not published and you have to contact sales
Does <code>ACT_GE_SCHEMA_LOG</code> match the actual schema in SIT, UAT and PROD?	Two Flyway migrations replayed the 7.20-to-7.22 upgrade by hand, guarded with <code>IF NOT EXISTS</code> . If the recorded version says below 7.22 while the columns already exist, the engine will fail to start at cutover . This is a strong inference, not yet verified	Four SQL queries per environment , run independently. Hours of work
How much does LORA not cover that Bravo does, per product and risk tier?	This is the largest uncertainty inside Option 3’s 30 to 57 engineer-months , and the input that tells you how much is new work rather than work already planned	A gap inventory, 1 to 2 engineer-months . Useful whichever option is chosen
How often do staff have to step in on Bravo? Answered: about 0.44%, against LORA’s 0.13% – 1 application in 225, against 1 in 790	No longer blocks the comparison. Reliability now favours LORA as clearly as workflow-tier cost favours Bravo	What is left is days of work : one ticket category – Surveyor Platform - Release reject, 1,542 tickets, 28.4% of the load, grown 4.8 times – accounts for the whole gap. Exclude it and Bravo is at 0.152%, level with LORA. Nobody has traced it to a BPMN element
How long does Bravo have to keep running?	It no longer picks a variant of Option 1. It now decides Decision B : whether to spend 30 to 57 engineer-months at all. That is a business decision, not a technical one	Steering committee. Set a horizon and publish it , even a provisional one

Fund Decision A now, and do not let it wait for Decision B

THE ONE FIRM RECOMMENDATION

Option 1, Path B — fork the engine and upgrade Spring Boot in one change.

18 to 33 engineer-days. No licence to buy. It ends with Bravo on a supported engine **and** a supported Spring Boot, rather than merely holding on to an unpatched one. Every hour of it is useful whichever strategic option is chosen, and it fixes an exposure that is live today.

1 Close the code-execution hole

Fix `SecurityConfig.java:65`, require authentication on the Camunda REST and web interfaces in production, delete the 61 injected definitions, and put the engine behind network isolation.
LIVE EXPOSURE · NEEDED WHICHEVER OPTION IS CHOSEN

2 Run the schema-log checks in every environment

Before committing to a cutover date.
This is the one finding that can change the plan.
HOURS · FOUR SQL QUERIES PER ENVIRONMENT

3 Do the six-item cleanup before deciding

Four of the six items are pure deletion. It is independent of both the path and the fork.
1–2 DAYS

4 Rehearse the cutover on a copy of production

And build the minimal parity test suite Bravo has never had at the engine layer.
Of `ms-bpm`'s 1,457 test files, 4 actually run a Camunda process.
THE MISSING SAFETY NET FOR ANY LATER CHANGE, INCLUDING BOTH MIGRATIONS

ONE PARTIAL SAFETY NET DOES EXIST, AND NOBODY HAD COUNTED IT UNTIL 2026-09-10

The three operator consoles carry **829 tests that gate their pull requests**, and those tests exercise the domain endpoints the engine sits behind. Because the console source contains **no Camunda references at all**, swapping the fork leaves them untouched. They can be run as a boundary smoke test.

Inside Option 1 one question is genuinely open, and it is not a technical one

THE QUESTION THAT PICKS THE FORK

Do we **need** a support contract for the workflow engine — and can one actually be bought for a BFI entity?

Yes → CIB seven. No → Operaton.

WHY IT IS ON THE CRITICAL PATH

It is the **only sub-decision we cannot answer from the codebase**, and CIB seven's terms — length, service levels, pricing — are **not published**. If the answer might be yes, open that enquiry *now*, rather than after the technical work has been scoped.

OPERATON 2.1.4

Nine times the commit volume, 52 contributors, Apache 2.0, with no paid tier. Maintained on rolling 6-month support lines. There is no legal entity behind it yet; a non-profit is planned.

Choose it if an active community matters more than a contract.

CIB SEVEN 2.2.0

Backed by a company (CIB software GmbH), with two ex-Camunda core engineers, and a cheaper migration — no need to go via 7.24 first, and the Cockpit webjar path is unchanged. Paid support exists, but **the terms are undisclosed**.

Choose it if Risk needs a counterparty.

EITHER WAY, THIS IS A DEPENDENCY WE ARE CHOOSING

Adopting a project with a 6-month support line will be **judged differently from drift we inherited**. Say so in the paper, rather than letting it come up in review.

The next 30 days, ordered so nothing here is wasted whichever option wins

#	ACTION	OWNER	WHY IT IS WORTH DOING REGARDLESS
1	Fix the Camunda code-execution hole. Rotate INTERNAL_SERVICE_KEY and the plaintext secrets in cypress.config.js. Identify who did it from the deployment access logs (2026-05-23 to 06-30). Register a ProcessEngineAuthenticationFilter. Delete the 61 definitions. Check whether anything moved laterally	Bravo and Security	Live, and code has already run in production. Needed whichever option is chosen
2	Run the schema-log checks in SIT, UAT and PROD independently – four SQL queries	Bravo	The one finding that can change the migration plan. Hours of work
2b	Do the six-item cleanup before deciding – swap CollectionUtil, delete instance-tab-modify.js, drop camunda-bpm-mockito and hibernate-types-55, remove the retired Jaeger stack and OldSecurityConfig.java	Bravo	1 to 2 days, and four of the six are pure deletion. Delaying it only makes the later diff messier
3	Set a Bravo horizon and publish it, even provisionally	Steering and CTO	The main input to Decision B
4	Answer the support-contract question – and if the answer might be yes, open the CIB seven enquiry now	Architect, procurement and Risk	It picks the fork. The critical path inside Option 1
5	Start the LORA gap inventory – map every Bravo delegate, status, assignment rule and approval path to an existing LORA ProcessStep, a new one that would be needed, or “drop it”	LORA and Bravo	Every number in Option 3 depends on it, and the per-product map is useful whichever option is chosen
6	Trace Surveyor Platform - Release reject to a BPMN element or an endpoint – there are now six named candidate endpoints	Bravo – squad LN	This one category accounts for 61% of the times Bravo needs a person. Narrowed from days of work to hours
7	Confirm how much Temporal headroom the contract has, ahead of the renewal around March 2027	Platform and FinOps	Needed for Option 2’s costing and Option 3’s capacity planning
8	Record less Camunda history on processes that are not audited	Bravo	full history kept for 90 days across 483 service tasks is a real slice of the Rp52.7M a month Cloud SQL line. Free money whichever option is chosen

How the estimates were made, and what we deliberately did not measure

HOW THE EFFORT IS COUNTED

Effort is in **engineer-months of a competent engineer who already knows the codebase**, including that engineer’s own testing but **not UAT run by business users**.

Option 1 is the exception. Its figures come from the Bravo team’s own *Camunda 7 Exit Plan*, in engineer-*days* for one engineer, and they explicitly exclude review, deployment soak and anything the open questions throw up — **18 to 33 days to build**, which we have converted here to about 1 to 1.5 engineer-months, or 1.5 to 2.5 all in. We re-checked their inventory counts against the working tree and they **matched exactly**. The day estimates carry the team’s own confidence rating: “**medium**”.

Ranges are **low to high**. They are not confidence intervals. Rupiah figures assume a **fully-loaded cost of Rp30–50M per engineer-month** — substitute BFI’s own rate card. USD converts at Rp16,800.

WHAT WE DID NOT MEASURE, AND WHAT IT WOULD CHANGE

The five open questions, plus how much headroom the Temporal contract has, plus the true split of applications between Bravo and LORA. **The billing sheet’s application counts are the least-verified numbers in the pack** — Bravo’s engine reports about 120,000 process starts a month, against the sheet’s 76,446. Option 3’s estimate also **does not separate new cost from work already on LORA’s roadmap**. The gap inventory is the piece of work that would.

SIZING INPUTS · COUNTED FROM MS-BPM @ 2D5D856

Main Java	497,970 lines / 5,070 files
Test files · of which run a Camunda process	1,457 · 4
Bravo LOS in total	4 repositories, 2,286 test files, plus 763,861 lines of console source
BPMN · DMN	53 · 3
delegateExpression bindings · beans	483 · 274
JavaDelegate implementations	225 (276 counting subclasses)
User tasks · call activities	96 · 56
Entities · tables · Flyway migrations	238 · 271 · 1,350
Active authors · commits in 2026	23 · 4,514

Two things to approve, and one thing to publish

1	Approve Decision A as spending that does not depend on anything else. Option 1, Path B — fork the engine and upgrade Spring Boot in one change, plus fixing the code-execution hole and running the schema-log checks. It commits you to nothing about Decision B.	18-33 ENG-DAYS RP45-125M NO LICENCE
2	Authorise the support-contract enquiry now, not later. Procurement and Risk answer one question: do we need a support contract for the workflow engine, and can one be bought for a BFI entity? That answer picks the fork.	WEEKS CRITICAL PATH INSIDE OPTION 1
3	Publish a Bravo horizon, even a provisional one. It is the main input to Decision B, and it is a business decision rather than a technical one. Without it, the 30-to-57 engineer-month options cannot be argued either way.	STEERING COMMITTEE
4	Commission the two measurements that would firm up Decision B — the LORA coverage gap inventory (1 to 2 engineer-months, useful whichever option is chosen) and tracing Release reject to its source (now hours, not days).	1-2 ENG-MONTHS + DAYS

On Decision B, this pack deliberately does not choose.

The three options answer different questions, and the coverage gap that would firm up Option 3’s number has not been measured. **What has changed is the pressure.** With Option 1 removing the end-of-support exposure in weeks rather than quarters, Decision B can be taken **on its merits** — architecture, way of building, one platform or two — instead of under a security deadline.